CIS 2267 Time-Series Data

Data Source Attribution and Citation

Source: US Bureau of Labor Statistics https://data.bls.gov/pdq/SurveyOutputServlet

All BLS data is in the public domain.

NOTE: All employment/unemployment numbers in this document are in thousands of people. In other words, add three zeros (000) to the end of each number.

Outliers

OUTLIERS are exceptionally big or small values compared to the rest of the data points. These anomalies (irregularities, one-offs) will affect (adversely?) any analysis you do. And thereby, might lead you to draw wrong conclusions. Anomalies might consist of missing datapoints, text in a numeric field, value exceeds allowable range, etc.

For examples of outliers, COVID affected almost everything during 2020 and 2021 (2020 more so than 2021, when recovery started, but still carrying into 2022). Look at the two charts below. The left chart is total employment; the right, total unemployment. COVID caused employment to plunge (left chart) and unemployment to spike (right). These outliers are not expected to recur regularly (we hope).

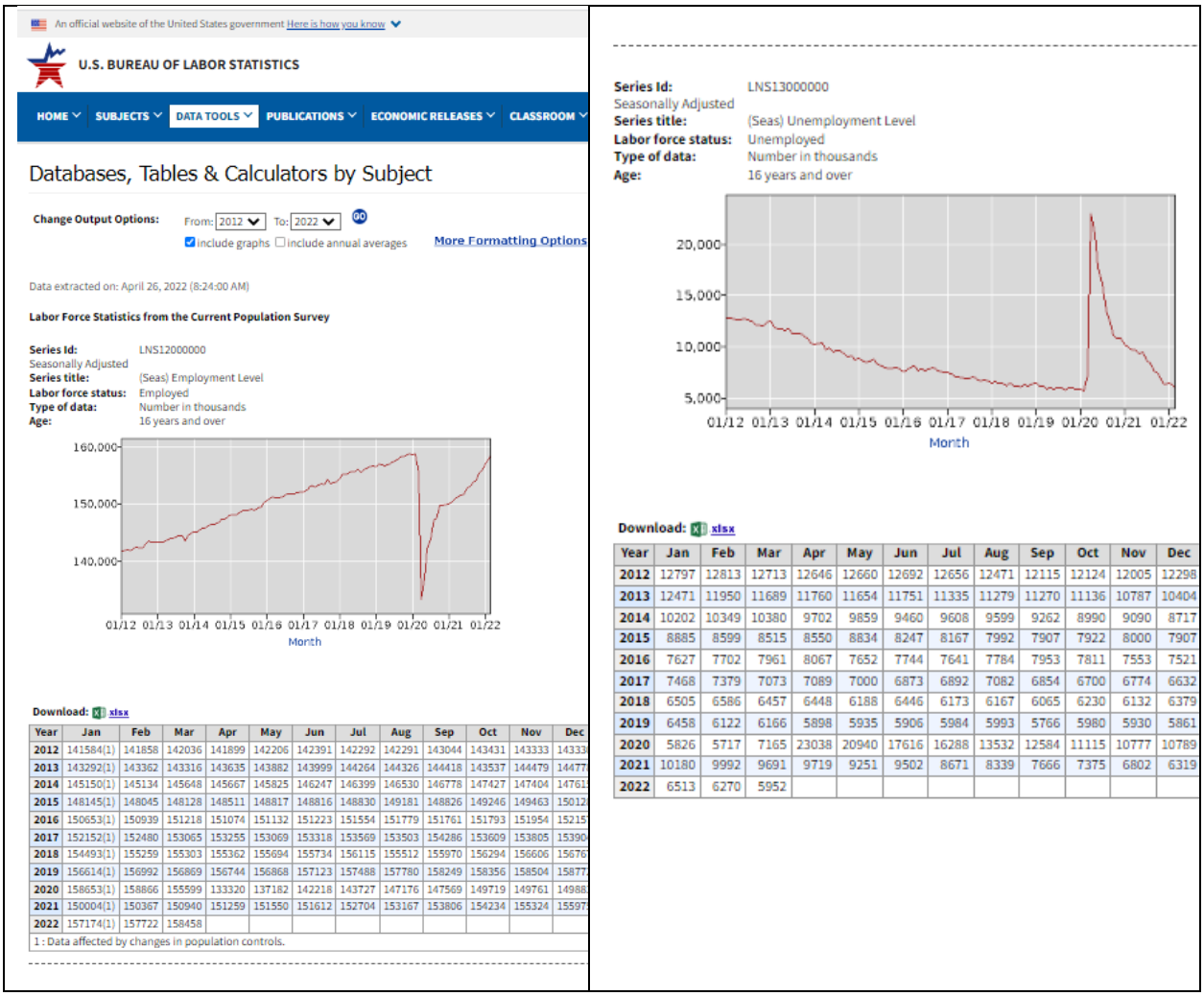
How should we handle those data points to not have undue influence on our analysis? That is the key question with all outliers.

As you can see from the below charts, aside from the COVID-induced spike and dip, the trend is upward, in the left employment chart, with almost full recovery from the dip by the end of the data, presumably to continue its upward trend in the future. Similarly, but opposite, for the right unemployment chart.

NOTE: It makes sense that employment and unemployment move in opposite directions. As employment rises, unemployment falls, and vice versa.

Discussion Questions: How best to handle the COVID-induced outliers?

- ✓ Would it make sense to simply eliminate the COVID datapoints?
- ✓ What are the consequences?
- ✓ Are there other alternatives?
- ✓ Which alternative for handling the outliers has the best tradeoffs of pros and cons?
- ✓ How would that be accomplished with Python?



COVID and global governments' decisions to lock down, affected everything in 2020 and 2021. Accordingly, whenever you are using data including those years, you must ask yourself how that data might have been affected and decide what, if anything, you should do to clean that data to remove or mitigate those effects.

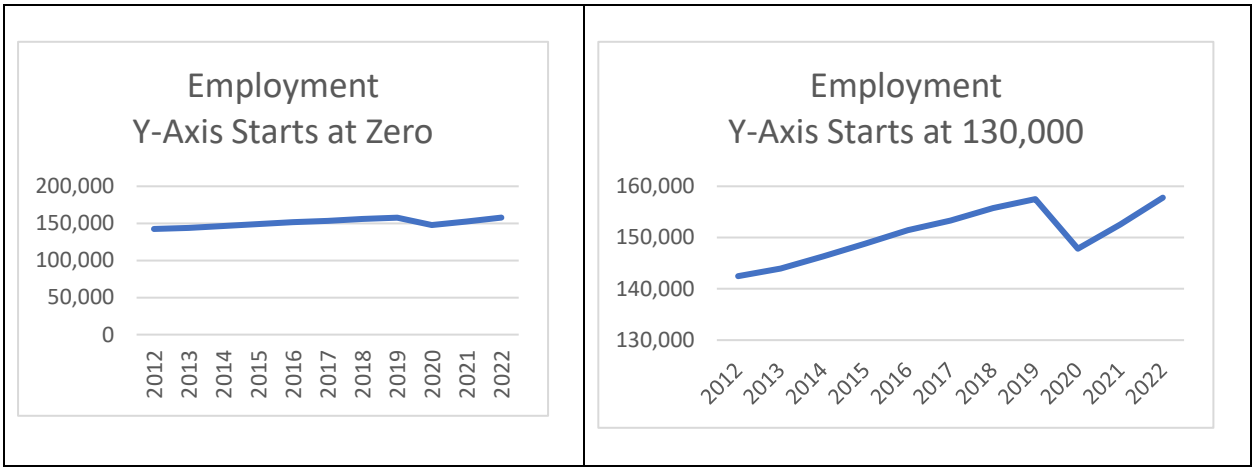
Dataset Summary

Notice in the employment/unemployment charts in the above screenshots, the data that generated the charts is listed below each chart. Notice also that it is exportable in Excel (xlsx) format. That data was used in the following analysis. All analysis in this document was done in Excel. To keep this manageable, we will use the employment data only. And we will *annualize* it to cut down the number of data points from 123 months to a more manageable 11 years + 3 months of 2022 (the most recent available as of this document publication date). Similar analysis could be made with the unemployment data.

Python has mechanisms to downsample data to accomplish a similar smoothing of the data.

Data Analysis

The first step in our analysis, notice how charts can be visually misleading. Below are two charts of the employment data. They plot the exact same data. They are both plotted correctly. But notice that the left chart COVID dip is hardly noticeable. That dip is very pronounced in the right chart. How can that be? Notice the Y-axis (vertical) in the left chart starts at zero, as viewers normally expect. That gives the visual impression that the dip is insignificant. The right chart's Y-axis starts at 130,000. This might have been done to conserve space or intentionally to mislead. Regardless of creator's intention, the right chart makes the dip more pronounced. The right chart corresponds to the left chart in the first BLS screenshot in this document: they both start at Y-axis = 130,000.



Data Cleaning

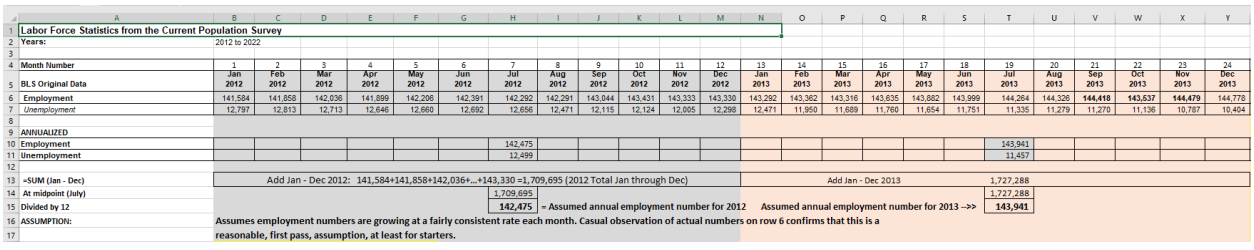
From now on, we will work with only the employment numbers. Unemployment numbers would be handled similarly.

1. Annualizing the Data:

The BLS data is by months. Annual numbers are not available. But we want annual numbers (for convenience and also to demonstrate this type of cleaning).

Assumptions: We make some assumptions whenever we clean data. The assumptions will either be reasonable or not and might be way off from reality, with consequences for all subsequent analysis.

Here, let's assume that each year's annual number is the sum of January through December, divided by 12, and placed in the middle of the year, July, as indicated in screenshot below.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1	Labor Force Statistics from the Current Population Survey																								
2	Years:	2012 to 2022																							
3																									
4	Month Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
5	BLS Original Data	Jan 2012	Feb 2012	Mar 2012	Apr 2012	May 2012	Jun 2012	Jul 2012	Aug 2012	Sep 2012	Oct 2012	Nov 2012	Dec 2012	Jan 2013	Feb 2013	Mar 2013	Apr 2013	May 2013	Jun 2013	Jul 2013	Aug 2013	Sep 2013	Oct 2013	Nov 2013	Dec 2013
6	Employment	141,584	141,858	142,036	141,899	142,206	142,391	142,292	142,291	143,044	143,431	143,333	143,330	143,292	143,362	143,316	143,635	143,882	143,999	144,264	144,326	**144,418**	**143,637**	**144,479**	144,778
7	*Unemployment*	12,797	12,813	12,713	12,646	12,660	12,692	12,656	12,471	12,115	12,124	12,095	12,296	12,471	11,950	11,689	11,760	11,654	11,751	11,335	11,279	11,270	11,136	10,787	10,484
8																									
9	ANNUALIZED																								
10	Employment							142,475												143,941					
11	Unemployment							12,499												11,457					
12																									
13	=SUM (Jan - Dec)	Add Jan - Dec 2012: 141,584+141,858+142,036+...+143,330 =1,709,695 (2012 Total Jan through Dec)												Add Jan - Dec 2013						1,727,288					
14	At midpoint (July)							1,709,695												1,727,288					
15	Divided by 12							**142,475**	= Assumed annual employment number for 2012					Assumed annual employment number for 2013 –>>						**143,941**					
16	ASSUMPTION:	Assumes employment numbers are growing at a fairly consistent rate each month. Casual observation of actual numbers on row 6 confirms that this is a																							
17		reasonable, first pass, assumption, at least for starters.																							

The annualizing process for 2022 is a bit different because data only exists for Jan – March. Accordingly, our formula will be to add Jan + Feb + March, divide by 3 months and center that annualized number on Feb, the middle month of these three months. We will assume that the centered, 3-month average for 2022 is also the full-year average number (but we know that that is not quite true because employment will (most likely) continue to grow April through December).

Repeating this averaging methodology for all years, yields the following assumed (cleaned) annual employment numbers. But effects of COVID outliers are still present (the yellow-highlighted 2020 and 2021 numbers below), to be removed later.

Screenshot of Annualized BLS Actual Numbers:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
27															
28															
29	YEAR NUMBER FOR REGRESSION EQUATION	1	2	3	4	5	6	7	8	9	10	11	12	13	14
30	2020 and 2021 are COVID-induced Outliers												TO BE FORECASTED		
31	ANNUALIZED *NOT* ADJUSTED for Outliers	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
32	Employment	142,475	143,941	146,319	148,845	151,436	153,335	155,759	157,530	147,806	152,579	157,785			
34															

2. Dealing with the Outliers

How to handle 2020 and 2021 outliers, if at all?

The method we will use is to assume that, instead of spiking and dipping as they did, the numbers continue to grow in 2020 and 2021 as they grew in the years prior to that. The mathematics of doing that follow:

Screenshot of Adjusting Methodology for 2020, 2021 Outliers:

	A	B	C	D	E	F	G	H	I	J	K
40	**COVID Adjustment Method Employment Data**										
41	1. Objective: You want the distance from 2019 to 2020 to be the same as the distance from 2020 to 2021										
42	to be the same distance from 2021 to 2022.										
43	2. Mathematically, this means divide the distance from 2019 to 2022 into 3 equal pieces, as follows:										
44	(2022 minus 2019)/3 = total difference divided into 3 pieces										
45	= (157,785 - 157530) / 3 = 255 / 3 = 85	The steps from 2019 to 2020 to 2021 to 2022 is 85 each year, as follows:									
46				2020	2021	2022					
47	2020 = 2019 + 1 piece of the 3 pieces = 157,530 + 85 =			157,615							
48	2021 = 2019 + 2 pieces = 2020 + 1 piece = 157,615 + 85 =				157,700						
49	2022 = 2019 + 3 pieces = 2021 + 1 piece = 157,700 + 85 =					157,785	<<-- should equal 2022 actual, and it does.				

Doing this produces the following results with the outliers smoothed away (i.e. their one-off severe dip is eliminated). Think about it: this method averages out the outliers, by spreading the differences between the non-outlier years (2019 and 2022) evenly between 2020 and 2021. By definition, any average, averages-out the underlying numbers, hiding that detail of how much the underlying numbers might have varied. For example, your average in this course is "C". That could mean that you were consistent, earning "C"s in all three exams or it could mean that you are improving (F, C, A) or barely surviving (A, C, F). They all average "C". Similarly here, our methodology averaged-away the big dip in 2020 2021 employment.

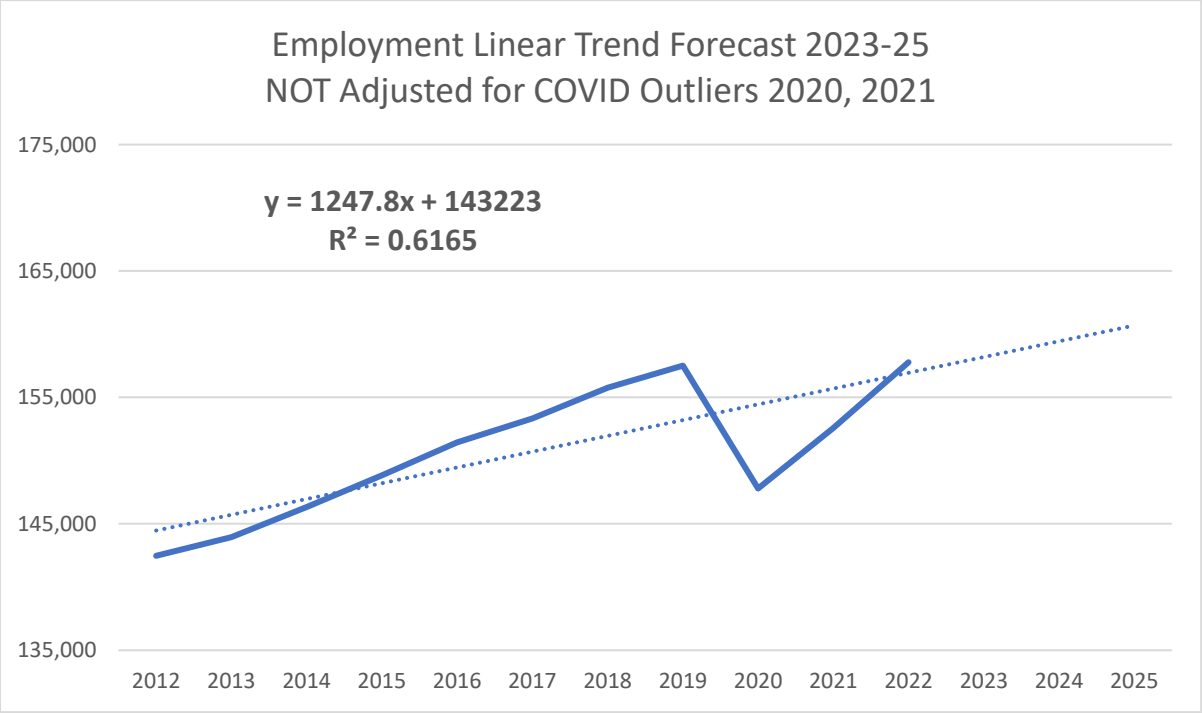
Screenshot of Annual Numbers Adjusted to Average Away the 2020 2021 COVID Outliers:

	A	B	C	D	E	F	G	H	I	J	K	L
34												
35												
36	**ANNUALIZED 2020, 21 ADJUSTED**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**
37	**Employment**	142,475	143,941	146,319	148,845	151,436	153,335	155,759	157,530	157,615	157,700	157,785
39												

The highlighted years have the COVID outlier effects removed. Compare this cleaned data with the actual numbers above. All the years are the same except 2020 and 2021, where we are now using outliers-adjusted data.

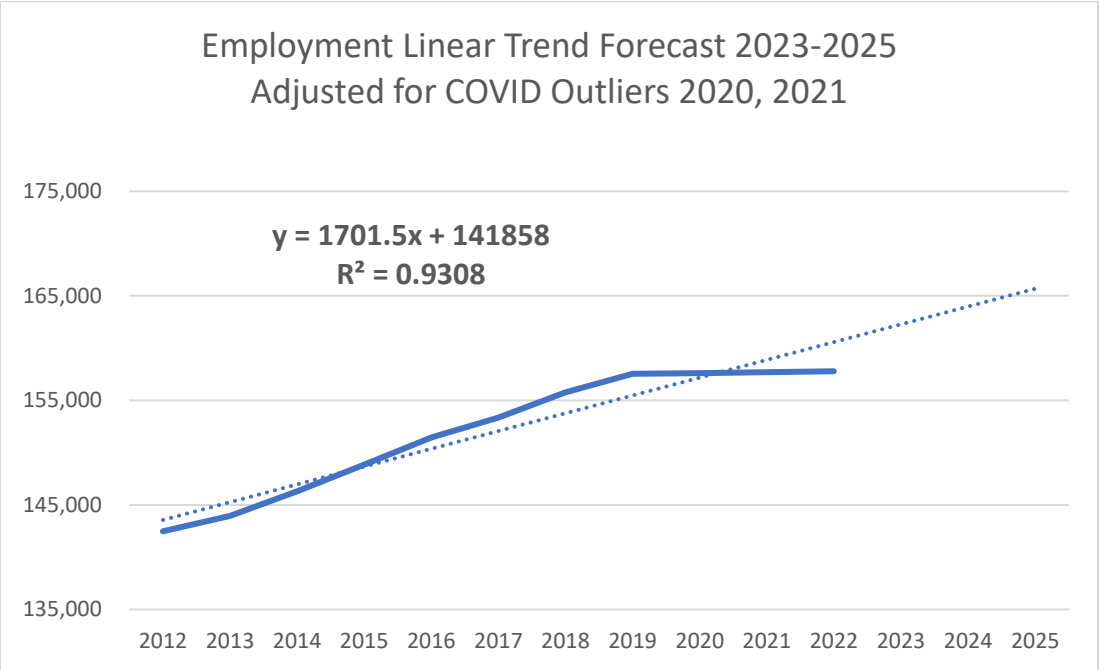
Case 0. Plotting a Linear Regression Line (Trendline) that Best Fits the Data Without Changing Outliers

Screenshot of Trendline Base Case 0: Not Adjusted for 2020 2021 Outliers:



Case 1. Plotting a Linear Regression Line (Trendline) that Best Fits the Data by Changing the Outliers

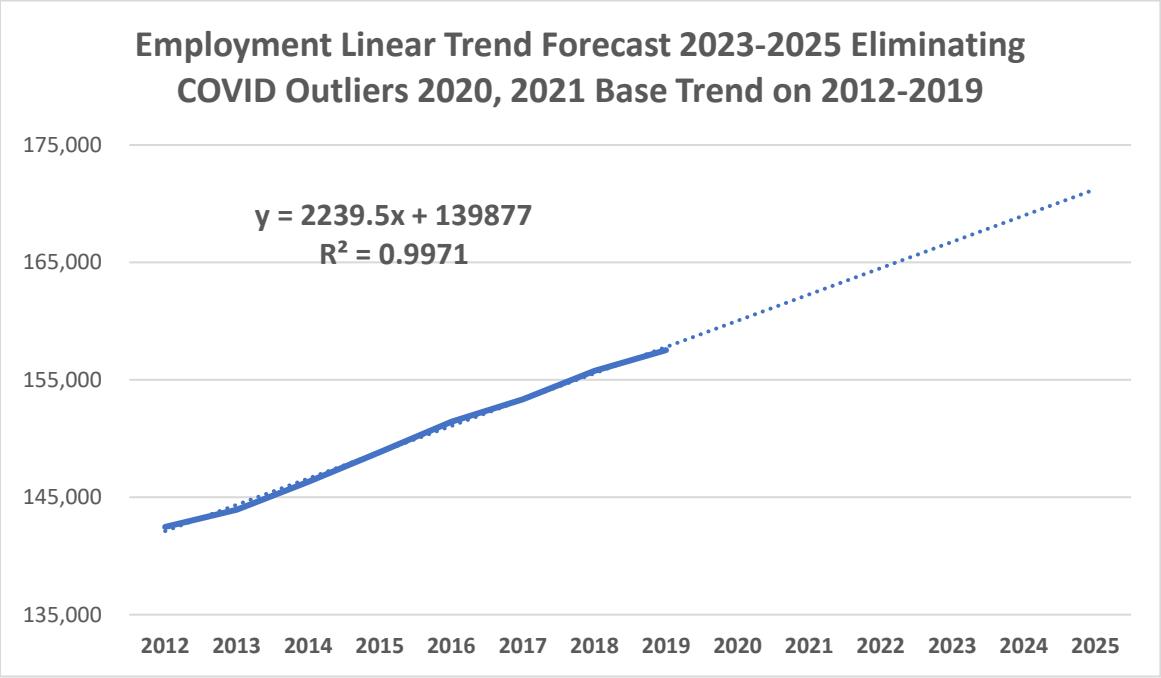
Screenshot of Trendline Case 1: 2020 2021 Outliers Adjusted by Averaging Methodology:



Case 2. Plotting a Linear Regression Line (Trendline) that Best Fits by Eliminating 2020 2021. Using 2012-2019 Only

As an alternative to the above methodology of averaging away the outliers, we could have simply eliminated 2020 and 2021 data. Accordingly, we could have produced our trendline using only pre-COVID years: 2012 through 2019. This was done below, producing a steeper trendline.

Screenshot of Trendline Case 2: 2020 2021 Outliers Eliminated. Trend based on 2012-2019 Data Only:



What is a Linear (Straight Line) Regression Line and its Equation (in the above 3 chart screenshots)?

Remembering your high school algebra, any straight line has the equation: Y = mX + b, where Y is the dependent variable (Employment), X is the independent variable (years), m is the slope (rise divided by run) of the straight line, and b is where the line cuts through (y-intercept) the Y-axis (vertical axis). Just to keep you confused, statistics textbooks change the letters and rearrange the terms but keep the same equation: Y = a + bX. The independent variable (years) is plotted on the horizontal axis; the dependent variable (employment) is plotted on the vertical axis.

What is R2 (in above 3 chart screenshots)?

A forecast (projection, prediction) is only as accurate (useful) as its R-squared (R^2) value (coefficient of determination). It measures the goodness of fit. R^2 can vary between zero (or 0%) and one (or 100%). Zero means that there is no correlation between X and Y. Accordingly, any forecasting based on this relationship is a waste of time. 100% means that there is perfect correlation: The points all line up exactly on a straight line. The bigger the R^2, the greater the correlation, the more useful our forecast is likely to be. We want a big R^2.

The above three charts show the equations and R^2 best-fit linear regression (trend) lines, under their respective assumptions on dealing with their outliers. There are mathematical formulas that compute these values. The math is beyond the scope of this course. Excel was used to compute these values automatically.

Cautions

1. Mathematically, you can make a forecast based on any trendline equation. The question is: How useful is that forecast likely to be based on your intended use? For example, you can forecast demand for your product 10 years into the future. But the farther out you forecast (say 10 years, rather than 1 or 2), the less reliable the forecast is likely to be. Are you willing to build a $10-billion factory based on this forecast? Because everything in the world is changing so quickly, organizations find that forecasting more than 1 or 2 or possibly 3 years into the future is not useful: things change too quickly.

2. Whenever you determine a trend, it is based on historical data. But the trend is subject to change at any time, as circumstances change. Theoretically, you should not use the regression line to forecast outside its relevant (historical) range. But with time series, it is pointless to try to predict the past. We want to predict the future, which by definition, is outside the relevant range. Therefore we risk the prediction being inappropriate because the trend changed.

3. The fewer years of historical data you base your trendline on, the less confidence you should have in the usefulness of the results.

4. A straight line (linear) might not be the best fit. Various curvilinear lines might fit better. Computers can try every permutation of variables and curve shapes to find the ones with the most explanatory power (biggest R^2).

Summarizing Results of Dealing With the 2020 2021 Outliers

The equations in the above three trendlines are based on years being the X-values. Accordingly, if you plug the years into the equations' X, and solve, you will have the forecasted employment levels (Y-values) in 2023, 2024, and 2025. BUT, mathematically, the way the formulas were derived, instead of using X = 2023, X = 2024, and X = 2025, these years are coded. Starting with 2012 = year 1, 2013 = 2, 2014 = 3….2022 = 11. Thus, you should plug X = 12 for 2023, X = 13 for 2024 and X = 14 for 2025 into the three equations and solve for Y (employment), as shown in the screenshot below.

What Sorts of Forecasts Do We Get Based on Varying Ways of Handling Outliers?

Remember our three cases, as discussed, and graphed above:

- Case 0 (Base Case): Make no adjustments for outliers
- Case 1: Average out 2020 and 2021 numbers
- Case 2: Eliminate 2020 2021 from consideration. Base trend on 2012 – 2019 data only

	A	B	C	D	E	F	G
101							
102					**Trend Forecast Using Trend (Regression) Equations**		
103	**2023-2025 Trend Forecasts:**	**Least-Squares Trend Equation**		**R^2**	**2023**	**2024**	**2025**
104	Year code for "x" for trend equation				12	13	14
105	Base Case 0:Unadjusted for COVID Outliers	y = 1247.8x + 143223		0.6165 or 61.65%	158,197	159,444	160,692
106	Case 1. Adjusted for COVID Outliers	y = 1701.5x + 141858		0.9308 or 93.08%	162,276	163,978	165,679
107	Case 2. Remove 2020-21. Base trendline on 2012-19	y = 2239.5x + 139877		0.9971 or 99.71%	166,751	168,991	171,230
108	**CLEANED MORE or (LESS) THAN UNADJUSTED**						
109	1. Adjusted for COVID Outliers				4,079	4,534	4,987
110	2. Remove 2020-21. Base trendline on 2012-19				8,554	9,547	10,538
111	**CLEANED MORE or (LESS) THAN UNADJUSTED (%)**						
112	1. Adjusted for COVID Outliers				2.6%	2.8%	3.1%
113	2. Remove 2020-21. Base trendline on 2012-19				5.4%	6.0%	6.6%
114							

Looking at the screenshot below:

- Row 104 is the year-codes to be substituted for X in the trendline equations. 2023=12; 2024=13; 2025=14
- Row 105-106 shows the equation's, R^2, and forecasted employment levels based on substituting X = 12; X=13, X=14 into the trend equations. Example: y=1247.8*12 + 143223 = 158,197 (forecast for 2023 for employment).
- Rows 109 110 show the greater number of people employed under Cases 1 and 2 than under the base case, Case 0. For example, cell G110 (red with yellow highlight), Case 2 employment in 2025 is 10,538,000 remember add 000 for thousands) more people employed than in Case 0. That's a lot more people.

Rows 112, 113 show how big is big? Absolute differences (rows 109, 110) do not convey how big is big? How small is small? That's because absolute differences do not make comparisons to other numbers. It is only by comparison with other numbers that we can begin to draw conclusions about how big is big. For example, Cell G113 (red with yellow highlight) = 6.6%. That's 6.6 squares compared to one-hundred squares…6.6% of anything is not particularly big.

We can see that the three approaches vary by 10½ million people in 2025, which is a lot of people, but as a proportion of the entire population of people employed (160-ish million), it's not an overly big difference. Government policy implications of these trends are beyond the scope of this course, but might be worth thinking about.

Your Tasks

Data for the above analysis has already been downloaded from the Bureau of Labor Statistics. It is available in both wide and long formats.

Develop and interactive Python notebook to perform similar analyses as described in this document. There is no required format or specific way to make your notebook. Instead, use Python and its associated libraries, your knowledge of math and statistics, and your own creativity.

Generally, you should follow this flow (though you are free to create your own):

1. Read the data from the file
2. Annualize the data, accounting for incomplete 2022 values
3. Create separate Pandas DataFrames for the three cases described above:
    a. Case 0: No adjustment for outliers
    b. Case 1: Average out the outliers
    c. Case 2: Eliminate outliers by using data before 2020
4. Draw plots of the three cases with trendlines
5. Calculate R^2 coefficients for the three cases
6. Forecast 2023, 2024, and 2025 employment for the three cases
7. Calculate the absolute difference (subtraction) and relative difference (percent) between Case 0 and Case 1
8. Use Markdown cells comments to describe your thought process and to document any code that might be difficult to understand
9. Use Markdown cells to create headings that organize your notebook